Investing in Data Science

Shaping the Present
&
Preparing for Growth

Barrett J. Viator
DSE5002-OM

Resolving Inefficiencies and Driving Data-driven Decision Making

Preparing our organization for future growth can best be accomplished by investing in data-driven decision making with the help of remote working specialists.

Initially, evaluations can be accomplished by a small number of dedicated staff.

As operations expand and more data is acquired, more members with advanced skills will need to be hired to utilize the data stream efficiently.

Recommendations for the Present

Who to hire...

Hiring a mid-level Data Scientist offshore to work on our project is a cost effective strategy.

A mid-level employee should be able to complete the necessary tasks without needing direct supervision.

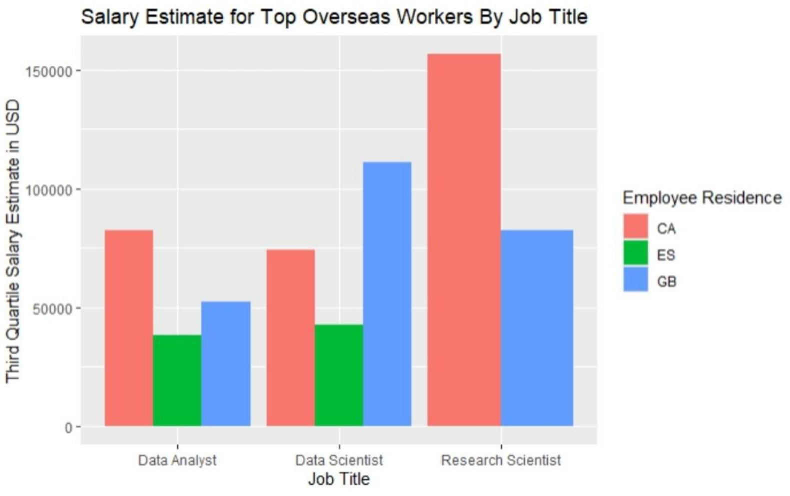
Expected Cost for Mid-Level Experience

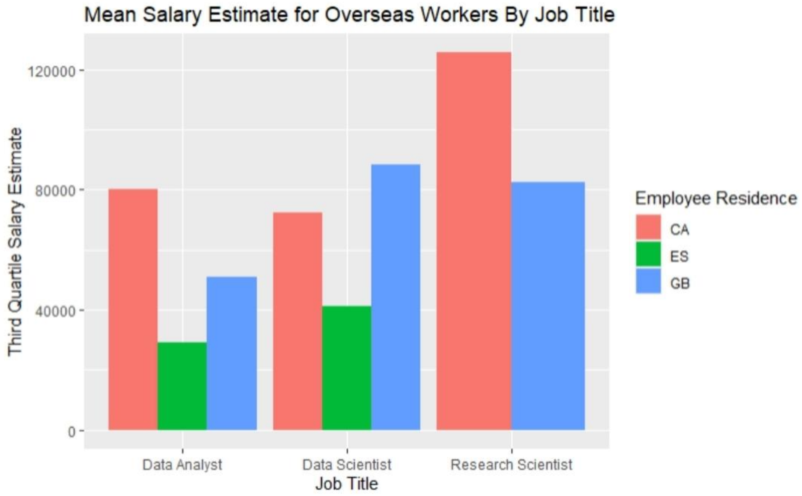
Offshore Data Scientist mean salaries are $77,404.

Top Data Scientist candidates are around $94,886.

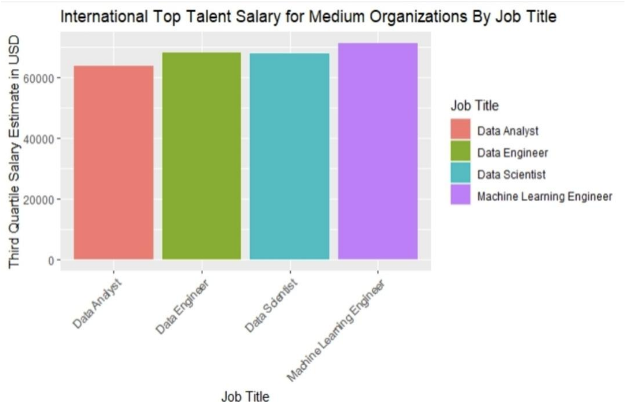
Data Analysts average at $50,167 with top candidates at $62,166.

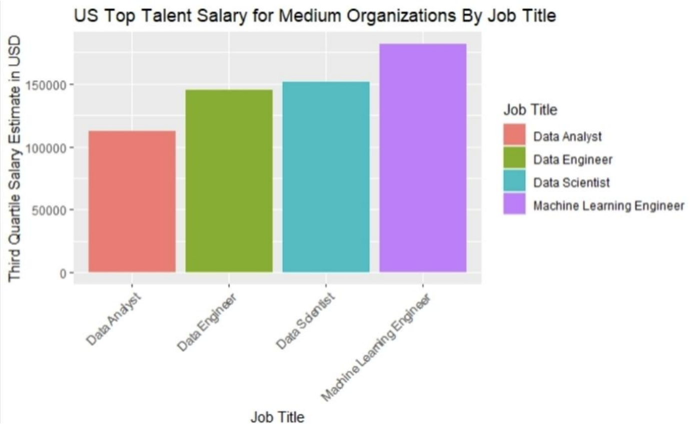
An average salary for a Research Scientist is $111,260, and top candidates are $134,985.





Expectations for Growth to Medium Corporation





As a medium size company, hiring offshore workers can still be an added benefit for saving on funds spent on overhead.

It may be more beneficial to look into remote US based workers who have a deeper understanding of the intricacies of the US market.

The team should be comprised of a Data Scientist, Data Engineer, Machine Learning Engineer, and a Data Science Manager.

The expected salaries for top employees are as follows, with US based first followed by oversees workers:

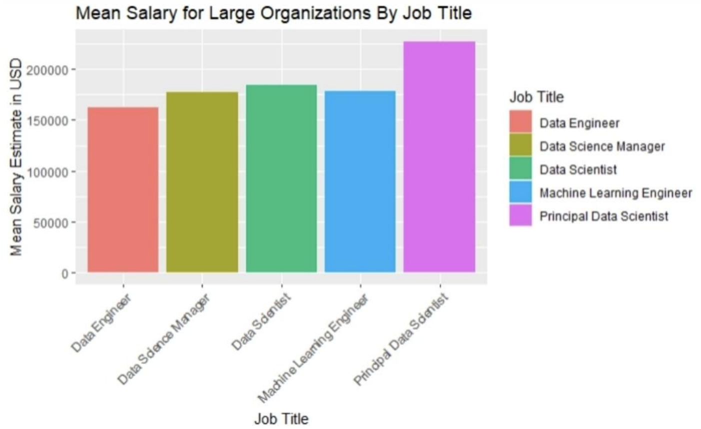
Data Analyst - $129,000 /$70,589

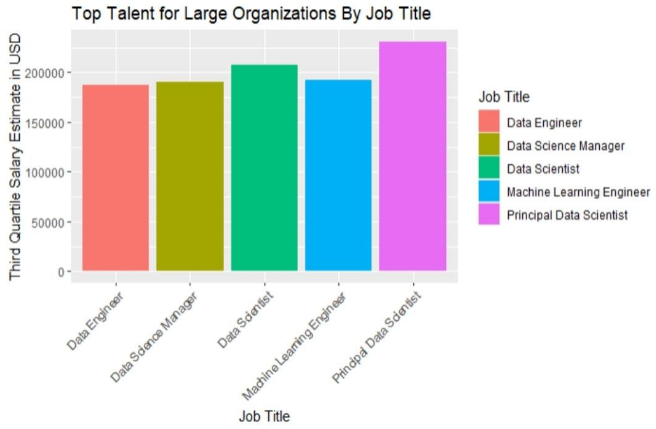
Data Scientist - $180,000/$90,386

Data Engineer - $167,000/$85,580

Machine Learning Engineer - $197,950/$91,030

Expectations for Growth to Large Corporation





The team will need to be comprised of a Data Scientist, Machine Learning Engineer, Data Science Manager, and a Head of Data Science.

Salary Expectations for top talent are as follows:

Data Scientist - $207,800

Machine Learning Engineer - $192,500

Data Science Manager - $190,500

Principal of Data Science - $231,250

Having a US-based team with top level talent will provide the organization with expertise to remain competitive in an ever evolving market.

Recap of Recommendations as Business Scales

Step 1

Analyze Current Position

Onboarding a skilled professional to gain initial insights and recommendations for next steps is suffice.

Following recommendations by further developing the team as business impacts are seen will be necessary.

Step 2

Refine Data Flow

Handling more data requires structuring the way data is collected.

A Data Engineer will need to be hired to manage these tasks.

Adding more members to the staff again will resolve issues brought by limitations of a small team.

Step 3

Optimize Team Efficiencies

Larger organizations require roles for better collaboration with executives.

This provides better support for team members while leveraging higher level skills when making more impactful decisions for long-term strategies.

DSE5002 - Project 1

Barrett Viator

2024-04-14

Process of Loading Data

```r
library(tidyr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)

salary_df <- read.csv("C:\\Program Files\\Merrimack\\DSE5002\\Project_1\\r_project_data.csv"
                      ,header = TRUE
                      ,stringsAsFactors = FALSE
                      )
```

```r
#Obtain a list of all of the job titles by level for insights
title_by_exp <- salary_df %>%
  group_by(experience_level, job_title) %>%
  count(job_title, na.rm = T, sort = TRUE)
  print(title_by_exp)
```

```
## # A tibble: 105 x 4
## # Groups:   experience_level, job_title [105]
##    experience_level job_title                 na.rm     n
##    <chr>            <chr>                     <lgl> <int>
##  1 SE               Data Engineer             TRUE     63
##  2 SE               Data Scientist            TRUE     61
##  3 MI               Data Scientist            TRUE     60
##  4 SE               Data Analyst              TRUE     54
##  5 MI               Data Engineer             TRUE     53
##  6 MI               Data Analyst              TRUE     29
##  7 EN               Data Scientist            TRUE     22
##  8 SE               Machine Learning Engineer TRUE     20
##  9 EN               Data Analyst              TRUE     12
## 10 EN               Data Engineer             TRUE     12
## # i 95 more rows
```

Steps taken to gather insights about counts

```r
#Obtain a list of all the experience levels with count for insights

count_of_exp_ <- salary_df %>%
  group_by(experience_level, job_title) %>%
  count(experience_level, sort = TRUE)
print(count_of_exp_)
```

```
## # A tibble: 105 x 3
## # Groups:   experience_level, job_title [105]
##    experience_level job_title                     n
##    <chr>            <chr>                     <int>
##  1 SE               Data Engineer                63
##  2 SE               Data Scientist               61
##  3 MI               Data Scientist               60
##  4 SE               Data Analyst                 54
##  5 MI               Data Engineer                53
##  6 MI               Data Analyst                 29
##  7 EN               Data Scientist               22
##  8 SE               Machine Learning Engineer    20
##  9 EN               Data Analyst                 12
## 10 EN               Data Engineer                12
## # i 95 more rows
```

```r
#Obtain a list for insights on salary and job titles for MI experience Level
count_of_exp_mi <- salary_df %>%
  filter(experience_level == "MI") %>%
  group_by(experience_level, job_title) %>%
  count(job_title, sort = TRUE)
print(count_of_exp_mi)
```

```
## # A tibble: 35 x 3
## # Groups:   experience_level, job_title [35]
##    experience_level job_title                              n
##    <chr>            <chr>                              <int>
##  1 MI               Data Scientist                        60
##  2 MI               Data Engineer                         53
##  3 MI               Data Analyst                          29
##  4 MI               Machine Learning Engineer             12
##  5 MI               Research Scientist                     7
##  6 MI               Machine Learning Scientist             4
##  7 MI               Applied Machine Learning Scientist     3
##  8 MI               BI Data Analyst                        3
##  9 MI               Big Data Engineer                      3
## 10 MI               Business Data Analyst                  3
## # i 25 more rows
```

```r
#Obtain a list for insights on salary and job titles for EN experience Level
count_of_exp_en <- salary_df %>%
  filter(experience_level == "EN") %>%
  group_by(experience_level, job_title) %>%
  count(job_title, sort = TRUE)
print(count_of_exp_en)
```

```
## # A tibble: 19 x 3
```

```
## # Groups:   experience_level, job_title [19]
##    experience_level job_title                              n
##    <chr>            <chr>                              <int>
##  1 EN               Data Scientist                        22
##  2 EN               Data Analyst                          12
##  3 EN               Data Engineer                         12
##  4 EN               Machine Learning Engineer              9
##  5 EN               Data Science Consultant                5
##  6 EN               AI Scientist                           4
##  7 EN               Research Scientist                     4
##  8 EN               Big Data Engineer                      3
##  9 EN               Computer Vision Engineer               3
## 10 EN               BI Data Analyst                        2
## 11 EN               Business Data Analyst                  2
## 12 EN               Computer Vision Software Engineer      2
## 13 EN               ML Engineer                            2
## 14 EN               Applied Data Scientist                 1
## 15 EN               Applied Machine Learning Scientist     1
## 16 EN               Data Analytics Engineer                1
## 17 EN               Financial Data Analyst                 1
## 18 EN               Machine Learning Developer             1
## 19 EN               Machine Learning Scientist             1
```

```r
#Obtain a list for insights on salary and job titles for EX experience Level
count_of_exp_ex <- salary_df %>%
  filter(experience_level == "EX") %>%
  group_by(experience_level, job_title) %>%
  count(job_title, sort = TRUE)
print(count_of_exp_ex)
```

```
## # A tibble: 13 x 3
## # Groups:   experience_level, job_title [13]
##    experience_level job_title                    n
##    <chr>            <chr>                    <int>
##  1 EX               Director of Data Science     6
##  2 EX               Data Engineer                4
##  3 EX               Head of Data Science         3
##  4 EX               Analytics Engineer           2
##  5 EX               Data Analyst                 2
##  6 EX               Head of Data                 2
##  7 EX               BI Data Analyst              1
##  8 EX               Data Engineering Manager     1
##  9 EX               Data Science Consultant      1
## 10 EX               Head of Machine Learning     1
## 11 EX               Lead Data Engineer           1
## 12 EX               Principal Data Engineer      1
## 13 EX               Principal Data Scientist     1
```

```r
#Obtain a list for insights on salary and job titles for SE experience Level
count_of_exp_se <- salary_df %>%
  filter(experience_level == "SE") %>%
  group_by(experience_level, job_title) %>%
  count(job_title, sort = TRUE)
print(count_of_exp_se)
```

```
## # A tibble: 38 x 3
```

```
## # Groups:   experience_level, job_title [38]
##    experience_level job_title                      n
##    <chr>            <chr>                      <int>
##  1 SE               Data Engineer                 63
##  2 SE               Data Scientist                61
##  3 SE               Data Analyst                  54
##  4 SE               Machine Learning Engineer     20
##  5 SE               Data Science Manager          10
##  6 SE               Data Architect                 8
##  7 SE               Data Analytics Manager         7
##  8 SE               Principal Data Scientist       5
##  9 SE               Research Scientist             5
## 10 SE               Lead Data Engineer             4
## # i 28 more rows
```

Steps taken to evaluate salaries based on residency

```
#Obtain a list of salary of US workers by small size US company
salary_list_s_us <- salary_df %>%
  filter(company_size == "S" & employee_residence == "US", employment_type == "FT") %>%
  group_by(experience_level,job_title, salary_in_usd)
print(salary_list_s_us)
```

```
## # A tibble: 20 x 12
## # Groups:   experience_level, job_title, salary_in_usd [19]
##        X work_year experience_level employment_type job_title             salary
##    <int>     <int> <chr>            <chr>           <chr>                  <int>
##  1     6      2020 SE               FT              Lead Data Scientist   190000
##  2    39      2020 EN               FT              Machine Learning Eng~ 138000
##  3    68      2020 EN               FT              Data Scientist        105000
##  4   107      2021 SE               FT              Data Engineer         115000
##  5   118      2021 EN               FT              Data Analyst           90000
##  6   126      2021 SE               FT              Machine Learning Sci~ 120000
##  7   134      2021 EN               FT              Data Scientist        100000
##  8   143      2021 MI               FT              Data Scientist         82500
##  9   159      2021 EN               FT              Machine Learning Eng~ 125000
## 10   168      2021 EN               FT              BI Data Analyst        55000
## 11   172      2021 EN               FT              Data Analyst           60000
## 12   178      2021 EN               FT              Machine Learning Eng~  81000
## 13   199      2021 EN               FT              Data Science Consult~  90000
## 14   231      2021 SE               FT              ML Engineer           256000
## 15   235      2021 MI               FT              Head of Data Science  110000
## 16   251      2021 EN               FT              Data Scientist         90000
## 17   367      2022 MI               FT              Data Analyst           58000
## 18   406      2022 MI               FT              Data Analyst           58000
## 19   453      2022 MI               FT              Machine Learning Eng~ 120000
## 20   512      2022 EN               FT              Data Engineer          65000
## # i 6 more variables: salary_currency <chr>, salary_in_usd <int>,
## #   employee_residence <chr>, remote_ratio <int>, company_location <chr>,
## #   company_size <chr>
```

```
#Obtain a list of salary of US workers by small size CA, ES, and GB companies for initial recommendatio
salary_list_s_offshore <- salary_df %>%
  filter(employment_type == "FT" &
```

```r
           experience_level == "MI" &
           employee_residence == "CA" |
           employee_residence == "ES" |
           employee_residence =="GB") %>%
  group_by(experience_level,job_title, salary_in_usd, employee_residence)
print(salary_list_s_offshore)
```

```
## # A tibble: 68 x 12
## # Groups:   experience_level, job_title, salary_in_usd, employee_residence [65]
##        X work_year experience_level employment_type job_title             salary
##    <int>     <int> <chr>            <chr>           <chr>                  <int>
##  1     2      2020 SE               FT              Big Data Engineer      85000
##  2    41      2020 EX               FT              Data Engineering Man~  70000
##  3    44      2020 MI               FT              Data Engineer          88000
##  4    46      2020 MI               FT              Data Scientist         60000
##  5    56      2020 MI               FT              Data Scientist         34000
##  6    61      2020 MI               FT              Data Engineer         130800
##  7    72      2021 EN               FT              Research Scientist     60000
##  8    82      2021 MI               FT              Applied Data Scienti~  68000
##  9    83      2021 MI               FT              Machine Learning Eng~  40000
## 10   105      2021 MI               FT              Data Analyst           37456
## # i 58 more rows
## # i 6 more variables: salary_currency <chr>, salary_in_usd <int>,
## #   employee_residence <chr>, remote_ratio <int>, company_location <chr>,
## #   company_size <chr>
```

```r
#Create observations for medium size companies in the US with FT employees
salary_list_m_us <- salary_df %>%
  filter(company_size == "M" & employee_residence == "US", employment_type == "FT") %>%
  group_by(experience_level,job_title, salary_in_usd)
print(salary_list_m_us)
```

```
## # A tibble: 210 x 12
## # Groups:   experience_level, job_title, salary_in_usd [170]
##        X work_year experience_level employment_type job_title             salary
##    <int>     <int> <chr>            <chr>           <chr>                  <int>
##  1    23      2020 MI               FT              BI Data Analyst        98000
##  2    33      2020 MI               FT              Research Scientist    450000
##  3    57      2020 MI               FT              Data Scientist        118000
##  4    59      2020 MI               FT              Data Scientist        138350
##  5    67      2020 SE               FT              Data Science Manager  190200
##  6    76      2021 MI               FT              BI Data Analyst       100000
##  7    79      2021 EN               FT              Data Analyst           80000
##  8    98      2021 EN               FT              Computer Vision Soft~  70000
##  9   108      2021 SE               FT              Data Engineer         150000
## 10   121      2021 SE               FT              Principal Data Engin~ 200000
## # i 200 more rows
## # i 6 more variables: salary_currency <chr>, salary_in_usd <int>,
## #   employee_residence <chr>, remote_ratio <int>, company_location <chr>,
## #   company_size <chr>
```

```r
#Create observations of M size businesses with overseas employees who work full time
salary_list_m_offshore <- salary_df %>%
  filter(company_size == "M" & employee_residence != "US", employment_type == "FT") %>%
  group_by(experience_level,job_title, salary_in_usd)
```

```r
print(salary_list_m_offshore)
```

```
## # A tibble: 108 x 12
## # Groups:   experience_level, job_title, salary_in_usd [96]
##        X work_year experience_level employment_type job_title              salary
##    <int>     <int> <chr>            <chr>           <chr>                   <int>
##  1     2      2020 SE               FT              Big Data Engineer       85000
##  2    12      2020 EN               FT              Data Scientist          35000
##  3    18      2020 EN               FT              Data Science Consult~ 423000
##  4    19      2020 MI               FT              Lead Data Engineer      56000
##  5    20      2020 MI               FT              Machine Learning Eng~ 299000
##  6    42      2020 MI               FT              Machine Learning Inf~  44000
##  7    55      2020 SE               FT              Principal Data Scien~ 130000
##  8    56      2020 MI               FT              Data Scientist          34000
##  9    61      2020 MI               FT              Data Engineer          130800
## 10    86      2021 EN               FT              Data Analyst            50000
## # i 98 more rows
## # i 6 more variables: salary_currency <chr>, salary_in_usd <int>,
## #   employee_residence <chr>, remote_ratio <int>, company_location <chr>,
## #   company_size <chr>
```

```r
#Gain insights about salaries for large companies with workers who live in the US
salary_list_l <- salary_df %>%
  filter(company_size == "L" & employee_residence == "US", employment_type == "FT") %>%
  group_by(experience_level,job_title, salary_in_usd)
print(salary_list_l)
```

```
## # A tibble: 98 x 12
## # Groups:   experience_level, job_title, salary_in_usd [91]
##        X work_year experience_level employment_type job_title              salary
##    <int>     <int> <chr>            <chr>           <chr>                   <int>
##  1     4      2020 SE               FT              Machine Learning Eng~ 150000
##  2     5      2020 EN               FT              Data Analyst            72000
##  3     8      2020 MI               FT              Business Data Analyst 135000
##  4    13      2020 MI               FT              Lead Data Analyst       87000
##  5    14      2020 MI               FT              Data Analyst            85000
##  6    25      2020 EX               FT              Director of Data Sci~ 325000
##  7    31      2020 EN               FT              Big Data Engineer       70000
##  8    36      2020 MI               FT              Data Science Consult~ 103000
##  9    37      2020 EN               FT              Machine Learning Eng~ 250000
## 10    43      2020 MI               FT              Data Engineer          106000
## # i 88 more rows
## # i 6 more variables: salary_currency <chr>, salary_in_usd <int>,
## #   employee_residence <chr>, remote_ratio <int>, company_location <chr>,
## #   company_size <chr>
```

```r
#Gather insights about salaries for employees residing outside of the US
salary_list_l_offshore <- salary_df %>%
  filter(company_size == "L" & employee_residence != "US", employment_type == "FT") %>%
  group_by(experience_level,job_title, salary_in_usd)
```

```r
#Investigate SE experience level US employees
salary_list_l_us <- salary_df %>%
  filter(company_size == "L" &
         employee_residence == "US",
```

```
        employment_type == "FT",
        experience_level == "SE") %>%
  group_by(experience_level,job_title, salary_in_usd)
```

##Steps taken to derive quartiles for salary expectations

```
#Look at Quartiles across different job titles and levels for narrowing down recommendations
job_title_quart <- salary_df %>%
  group_by(job_title)%>%
 summarise(M=mean(salary_in_usd),
          Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```
#Look at Quartiles of large companies by job title in the US
job_title_quart_l_us <- salary_list_l_us %>%
  group_by(job_title)%>%
 summarise(M=mean(salary_in_usd),
          Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```
# Look at Quartiles of salaries of large companies based outside of the US

job_title_quart_l_offshore <- salary_list_l_offshore %>%
  group_by(job_title)%>%
  arrange(salary_list_l_offshore, experience_level) %>%
 summarise(M=mean(salary_in_usd),
          Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```
job_title_quart_m_us <- salary_list_m_us %>%
  group_by(job_title)%>%
  arrange(salary_list_m_us, experience_level) %>%
 summarise(M=mean(salary_in_usd), Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```
job_title_quart_m_offshore <- salary_list_m_offshore %>%
  filter( job_title == "Data Scientist"|
           job_title == "Machine Learning Engineer" |
           job_title == "Data Analyst" |
           job_title == "Data Engineer" ) %>%
  group_by(job_title)%>%
  summarise(M=mean(salary_in_usd), Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```
job_title_quart_s_us <- salary_list_s_us %>%
  group_by(job_title)%>%
```

```r
  arrange(salary_list_s_us, experience_level) %>%
 summarise(M=mean(salary_in_usd),
          Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```r
job_title_quart_s_offshore <- salary_list_s_offshore %>%
  filter(job_title == "Data Analyst" |
          job_title == "Data Scientist" |
          job_title == "Research Scientist") %>%
  group_by(job_title, employee_residence)%>%
  #arrange(job_title) %>%
 summarise(M=mean(salary_in_usd),
          Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```
## `summarise()` has grouped output by 'job_title'. You can override using the
## `.groups` argument.
```

```r
#Get total quartiles of salary estimates by job title in selected overseas countries
job_title_quart_s_offshore_total <- salary_list_s_offshore %>%
  filter(job_title == "Data Analyst" |
          job_title == "Data Scientist" |
          job_title == "Research Scientist") %>%
  group_by(job_title, employee_residence)%>%
 summarise(M=mean(salary_in_usd),
          Med=median(salary_in_usd),
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))
```

```
## `summarise()` has grouped output by 'job_title'. You can override using the
## `.groups` argument.
```

```r
#Create table for insights into hiring offshore workers from multiple sized companies with MI experienc
salary_list_offshore <- salary_df %>%
  filter(company_size == "S" |
          company_size == "M" |
          company_size == "L" &
          employee_residence != "US",
        employment_type == "FT",
        experience_level == "MI") %>%
         group_by(experience_level,job_title, salary_in_usd)

#Create table of quartiles based on the job titles of positions recommended
job_title_quart_offshore <- salary_list_offshore %>%
  filter(job_title == "Data Analyst" |
          job_title == "Data Scientist" |
          job_title == "Research Scientist") %>%
  group_by(job_title)%>%
 summarise(M=mean(salary_in_usd),
          Med=median(salary_in_usd),
```

```r
          Q1=quantile (salary_in_usd, probs=0.25),
          Q2=quantile (salary_in_usd, probs=0.50),
          Q3=quantile(salary_in_usd, probs=0.75))

#Rename data frame for further analysis
recommendation_s_offshore <- salary_list_s_offshore %>%
  filter(job_title == "Data Analyst" |
          job_title == "Data Scientist" |
          job_title == "Research Scientist")

print(recommendation_s_offshore)
```

```
## # A tibble: 29 x 12
## # Groups:   experience_level, job_title, salary_in_usd, employee_residence [29]
##        X work_year experience_level employment_type job_title          salary
##    <int>     <int> <chr>            <chr>           <chr>               <int>
##  1    46      2020 MI               FT              Data Scientist      60000
##  2    56      2020 MI               FT              Data Scientist      34000
##  3    72      2021 EN               FT              Research Scientist  60000
##  4   105      2021 MI               FT              Data Analyst        37456
##  5   106      2021 MI               FT              Research Scientist 235000
##  6   124      2021 EN               PT              Data Analyst         8760
##  7   152      2021 MI               FT              Data Scientist      95000
##  8   202      2021 MI               FT              Data Scientist      32000
##  9   221      2021 MI               FT              Data Scientist      85000
## 10   223      2021 MI               FT              Data Scientist      40900
## # i 19 more rows
## # i 6 more variables: salary_currency <chr>, salary_in_usd <int>,
## #   employee_residence <chr>, remote_ratio <int>, company_location <chr>,
## #   company_size <chr>
```

```r
# Produce data frame for final result for large company US recommendation
job_title_quart_l_us_final <- job_title_quart_l_us %>%
  filter( job_title == "Data Scientist"|
            job_title == "Machine Learning Engineer" |
            job_title == "Data Science Manager" |
            job_title == "Principal Data Scientist" |
            job_title == "Data Engineer") %>%
  group_by(job_title)
```

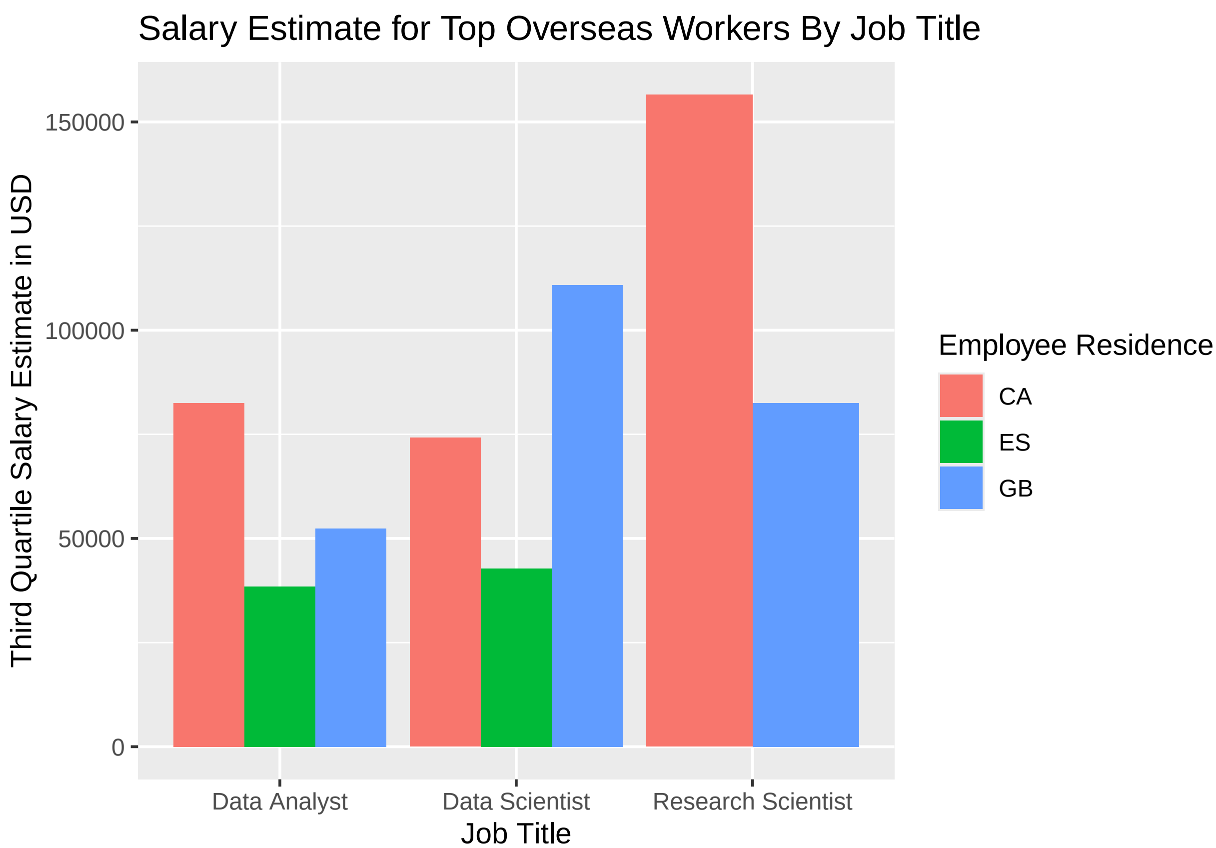
```r
# Produce data frame for final results of US medium size company recommendations
job_title_quart_m_us_final <- job_title_quart_m_us %>%
  filter( job_title == "Data Scientist"|
            job_title == "Machine Learning Engineer" |
            job_title == "Data Analyst" |
            job_title == "Data Engineer" ) %>%
  group_by(job_title)
```

```r
# Produce data frame for final results of US medium size company recommendations with overseas workers
job_title_quart_m_offshore_final <- job_title_quart_m_offshore %>%
  filter( job_title == "Data Scientist"|
            job_title == "Machine Learning Engineer" |
            job_title == "Data Analyst" |
            job_title == "Data Engineer" ) %>%
  group_by(job_title)
```

Included Plots

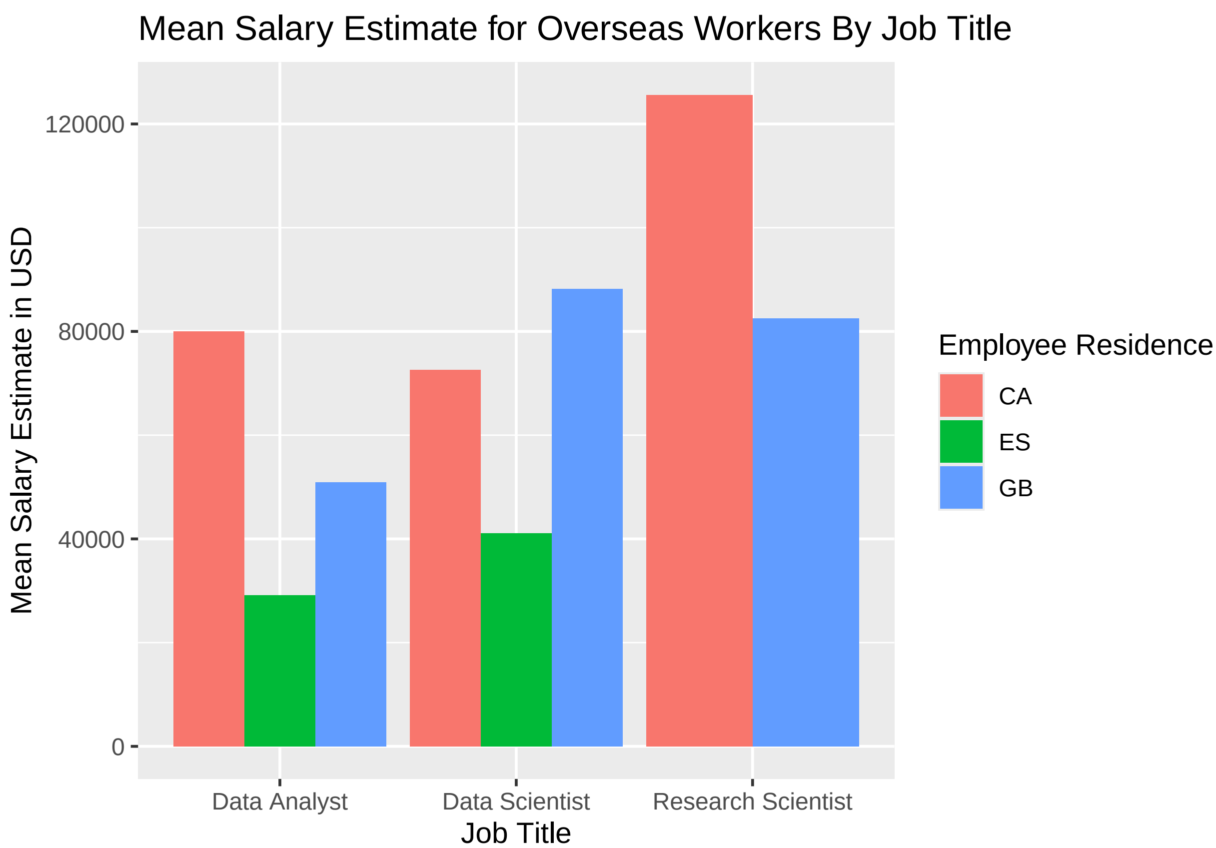
```
#create bar plot of salary expectations for top employees for initial recommendation
ggplot(job_title_quart_s_offshore_total, aes(fill=employee_residence, x= job_title, y = Q3)) +
  geom_col(position ="dodge", stat = "identity") +
  ggtitle("Salary Estimate for Top Overseas Workers By Job Title") +
  xlab("Job Title") +
  ylab("Third Quartile Salary Estimate in USD") +
  labs(fill = "Employee Residence")
```

```
## Warning in geom_col(position = "dodge", stat = "identity"): Ignoring unknown
## parameters: `stat`
```



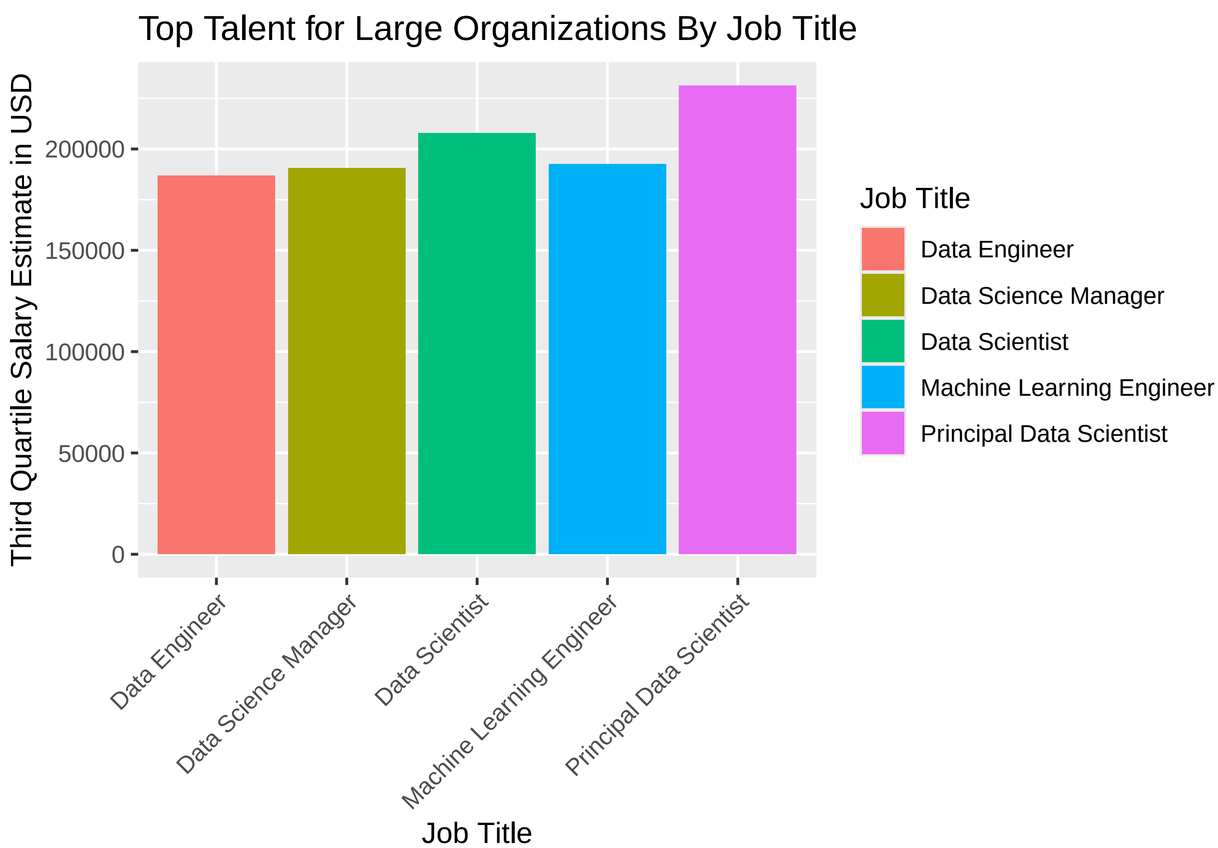

```
#create bar plot of salary expectations for average employees initial recommendation
ggplot(job_title_quart_s_offshore, aes(fill=employee_residence, x= job_title, y = M)) +
  geom_col(position ="dodge", stat = "identity") +
  ggtitle("Mean Salary Estimate for Overseas Workers By Job Title") +
  xlab("Job Title") +
  ylab("Mean Salary Estimate in USD") +
  labs(fill = "Employee Residence")
```

```
## Warning in geom_col(position = "dodge", stat = "identity"): Ignoring unknown
## parameters: `stat`
```




```
#create bar plot for recommendation of roles with third quartile salary for US large company
ggplot(job_title_quart_l_us_final, aes(fill = job_title, x= job_title, y = Q3)) +
  geom_col(position ="dodge", stat = "identity") +
  ggtitle("Top Talent for Large Organizations By Job Title") +
  xlab("Job Title") +
  ylab("Third Quartile Salary Estimate in USD") +
  labs(fill = "Job Title") +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1))
```

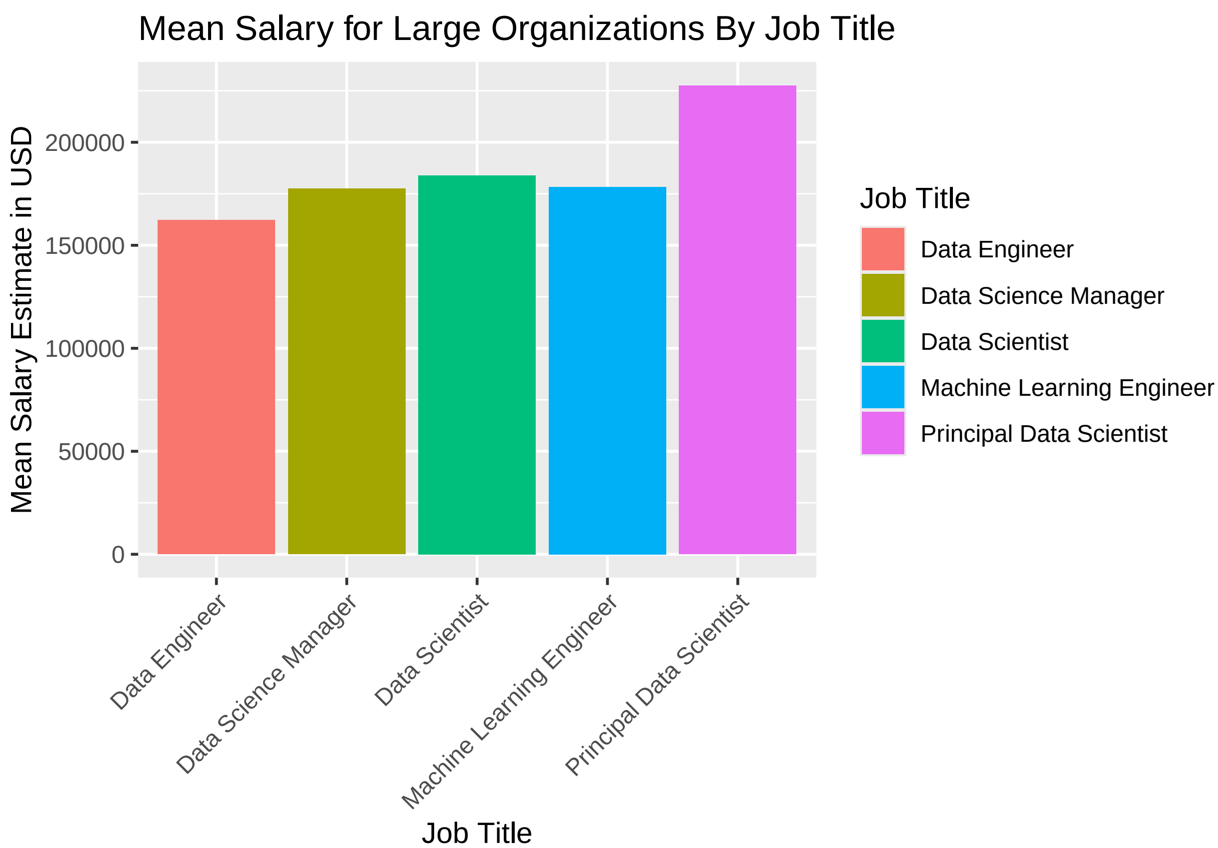
```
## Warning in geom_col(position = "dodge", stat = "identity"): Ignoring unknown
## parameters: `stat`
```




```r
#create bar plot for recommendation of roles with mean salary for US large company

ggplot(job_title_quart_l_us_final, aes(fill = job_title, x= job_title, y = M)) +
  geom_col(position ="dodge", stat = "identity") +
  ggtitle("Mean Salary for Large Organizations By Job Title") +
  xlab("Job Title") +
  ylab("Mean Salary Estimate in USD") +
  labs(fill = "Job Title") +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1))
```

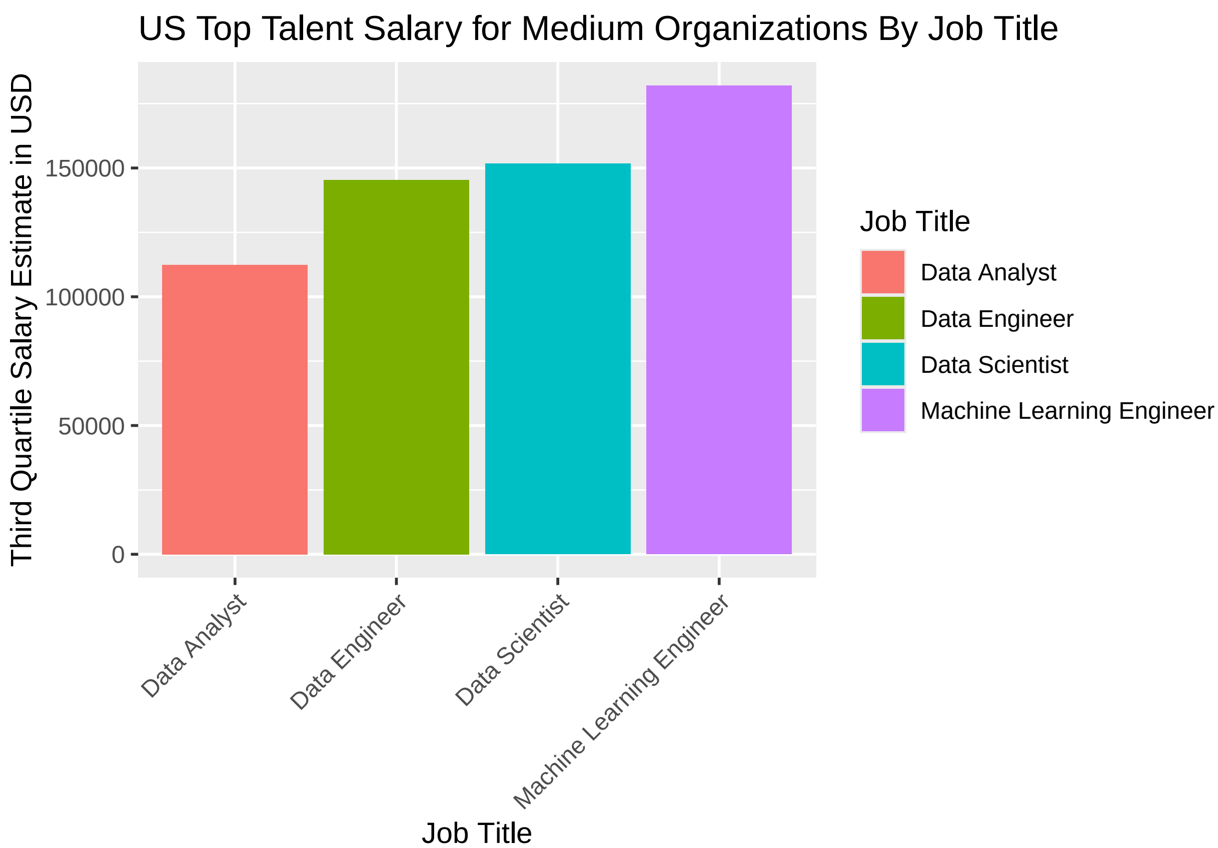
```
## Warning in geom_col(position = "dodge", stat = "identity"): Ignoring unknown
## parameters: `stat`
```




```r
#create bar plot for recommendation of roles with third quartile salary for US large company

ggplot(job_title_quart_m_us_final, aes(fill = job_title, x= job_title, y = M)) +
  geom_col(position ="dodge", stat = "identity") +
  ggtitle("US Top Talent Salary for Medium Organizations By Job Title") +
  xlab("Job Title") +
  ylab("Third Quartile Salary Estimate in USD") +
  labs(fill = "Job Title") +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1))
```

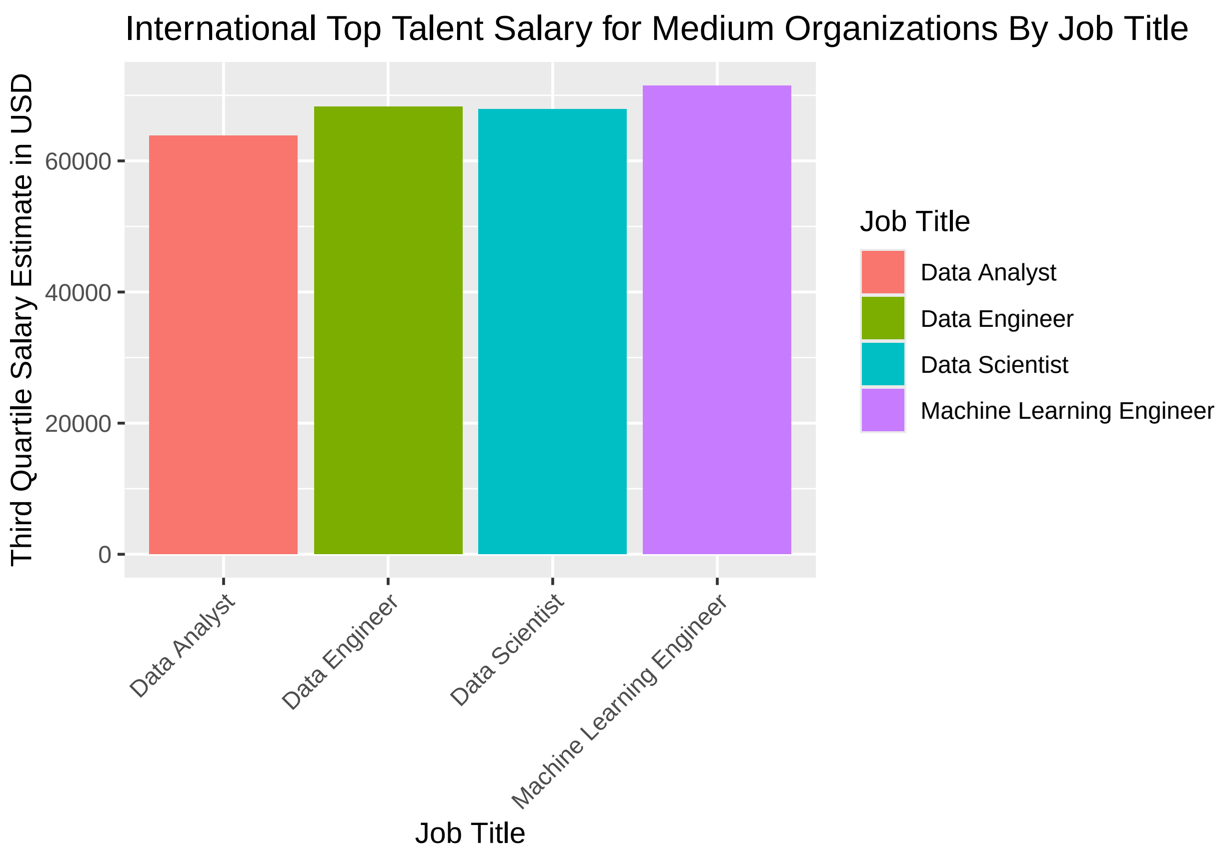
```
## Warning in geom_col(position = "dodge", stat = "identity"): Ignoring unknown
## parameters: `stat`
```




```r
#create bar plot for recommendation of roles with third quartile salary for US large company with overs

ggplot(job_title_quart_m_offshore_final, aes(fill = job_title, x= job_title, y = M)) +
  geom_col(position ="dodge", stat = "identity") +
  ggtitle("International Top Talent Salary for Medium Organizations By Job Title") +
  xlab("Job Title") +
  ylab("Third Quartile Salary Estimate in USD") +
  labs(fill = "Job Title") +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1))
```

```
## Warning in geom_col(position = "dodge", stat = "identity"): Ignoring unknown
## parameters: `stat`
```


International Top Talent Salary for Medium Organizations By Job Title
Third Quartile Salary Estimate in USD
60000
40000
20000
0
Data Analyst
Data Engineer
Data Scientist
Machine Learning Engineer
Job Title
Job Title
Data Analyst
Data Engineer
Data Scientist
Machine Learning Engineer